

TEIT 2171 – Computer Networks II (2 Credits)

Course Description

Computer Networks II examines network design and architecture considerations, network documentation, change management, network monitoring methods and solutions, configuration management, network security hardening techniques, and basic datacenter, cloud, and virtual-network concepts. This course aligns with objectives from popular networking certifications.

Course Objectives

- Compare and contrast networking appliances, applications, and functions.
- Identify the purpose of organizational processes and procedures.
- Use protocols, tools, and techniques to monitor network activity and troubleshoot performance and availability issues.
- Identify and implement network defense techniques, security features, and security solutions.
- Configure secure enterprise wireless networks.
- Identify basic datacenter, cloud, and virtual-networking concepts.
- Compare and contrast network access and management methods.

Course Outline

- Specialized Networks
- Wireless Networking
- Wide Area Networks, Operations and Management
- Security, Optimization and Troubleshooting

Textbook & Reading Materials

CompTIA Network+ CertMaster Learn (ISBN: 9781642745245)

Assignments & Assessments

10.1 - Authentication	12.2.8 - Lab: Design an Indoor Wireless Network
10.1.9 - Lesson Review	12.2.9 - Lab: Design an Outdoor Wireless Network
10.2.5 - Lab: Manage Account Policies	12.2.10 - Lab: Implement an Enterprise Wireless Network
10.2.6 - Lesson Review	12.2.11 - Lesson Review
10.3.3 - View Linux Services	12.3 - Wireless Security
10.3.5 - Lab: Scan for Unsecure Protocols	12.3.7 - Lab: Configure a Captive Portal
10.3.6 - Lab: Enable and Disable Linux Services	12.3.8 - Lab: Create a Guest Network for BYOD
10.3.7 - Lab: Disable Network Service	12.3.9 - Lab: Secure an Enterprise Wireless Network
10.3.8 - Lesson Review	12.3.10 - Lab: Secure a Home Wireless Network
10.4.2 - Lab: Secure Access to a Switch	12.3.11 - Lab: Enable Wireless Intrusion Prevention
10.4.3 - Lab: Secure Access to a Switch 2	12.3.12 - Lesson Review
10.4.4 - Lab: Disable Switch Ports - GUI	12.4.7 - Lab: Explore Wireless Network Problems
10.4.7 - Lab: Harden a Switch	12.4.8 - Lab: Troubleshoot Wireless Network Problems
10.4.9 - Lesson Review	12.4.9 - Lab: Optimize a Wireless Network
Student Acknowledgement Statement Module 1	12.4.10 - Lesson Review
Checkpoint Meeting Module 1	12.5 - Module Quiz
10.5 - Network Security Rules	12.6 - Checkpoint Review
10.5.6 - Lab: Configure Network Security Appliance Access	Student Acknowledgement Statement Module 3
10.5.7 - Lab: Configure a Security Appliance	Checkpoint Meeting Module 3
10.5.8 - Lab: Configure a Perimeter Firewall	13.1.4 - Lesson Review
10.5.9 - Lab: Restrict Telnet and SSH Access	13.2.8 - Lab: Configure a Remote Access VPN
10.5.10 - Lab: Permit Traffic	13.2.9 - Lab: Configure an iPad VPN Connection
10.5.11 - Lab: Block Source Hosts	13.2.10 - Lab: Configure a RADIUS Solution
10.5.12 - Lesson Review	13.2.11 - Lesson Review
10.6 - Module Quiz	13.3.8 - Lab: Allow Remote Desktop Connections
11.1.5 - Lab: Configure a Screened Subnet (DMZ)	13.3.9 - Lab: Use PowerShell Remote
11.1.6 - Lab: Configure Screened Subnets	13.3.10 - Lesson Review
11.1.9 - Lab: Implement Intrusion Prevention	13.4 - Module Quiz
11.1.10 - Lesson Review	14.1.5 - Lab: Configure an iSCSI Target
11.2.5 - Lab: Scan for IoT Devices	14.1.6 - Lab: Configure an iSCSI Initiator
11.2.6 - Lesson Review	14.1.7 - Lesson Review
11.3.4 - Lab: Implement Physical Security	14.2.5 - Lesson Review
11.3.5 - Lesson Review	14.3.7 - Lesson Review
11.4 - Module Quiz	14.4.9 - Lesson Review
12.1.9 - Lab: Configure Wireless Profiles	14.5 - Module Quiz
12.1.10 - Lesson Review	Student Feedback Survey
Student Acknowledgement Statement Module 2	End of Course Student Acknowledgement Statement
Checkpoint Meeting Module 2	Course Completion Checkpoint Meeting

Subject to change. Please consult your Canvas course for the most current instructions and updates.

Classroom Hours

Mo, Tu, W, Th, Fr

8:00 AM - 11:00 AM

12:00 PM - 3:00 PM

For a full list of course hours visit: [Course Schedule](#)

Instructor Contact Information

Greg Davis — gdavis@stech.edu

Austin Prince — aprince@stech.edu

Office Hours: By appointment

Email is the preferred method of communication; you will receive a response within 24 hours during regular business hours.

Canvas Information

Canvas is where course content, grades, and communication will reside for this course.

- Canvas URL: stech.instructure.com
 - For Canvas passwords or any other computer-related technical support contact Student Services.
 - For Regular Hours and Weekdays call (435) 586 - 2899
 - For After Hours and Weekends call (435) 865 - 3929 (Leave a message if no response)
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Policies

Course Grading

All assignments in this course require 100% score with unlimited submissions. All quizzes require a minimum score of 80%.

High School Power School Grades: Quarter student grades will be determined by student progress percentage. Faculty will use the higher percentage of either 1) quarter progress, or 2) cumulative progress for the current training plan year.

Grade Scale: The following grading scale will be used to determine a letter grade.

A : 94 - 100%

A-: 90 - 93%

B+: 87 - 89%

B : 83 - 86%

B-: 80 - 82%

C+: 77 - 79%

C : 73 - 76%

C-: 70 - 72%

D+: 67 - 69%

D : 63 - 66%

D-: 60 - 62%

F : 0 - 59%

Course Policies

You are required to keep your progress and attendance at 67% minimum. You must complete this program within 150% estimated program length. You are permitted one 15 minute break every 90 minutes. If you take more than one break in a 90 minute period or your break lasts longer than 15 minutes, your attendance will be penalized. 10 consecutive absences will lead to being withdrawn from the program. Please notify your instructors about absences as soon as possible. If absence is due to illness, please email your instructors prior to end of day. Cell Phone/Electronics – Cell phones cannot be used during class time. You may bring your personal computers to class. You must be on topic in the lab while clocked-in. Industry Environment – computer science typically is very sedentary. This means you may sit at a desk for long hours. Be sure to move and get what exercise you can.

Additional Information

InformaCast Statement: Southwest Tech uses InformaCast to ensure the safety and well-being of our students. In times of emergency, such as weather closures and delays, this app allows us to promptly deliver notifications directly to your mobile devices. To stay informed and receive real-time updates, we encourage all students to sign up for notifications. Your safety is our priority, and staying connected ensures a swift response to any unforeseen circumstances. More information and directions for signing up are available at: <https://stech.edu/emergency-notifications/>

Internet Acceptable Use Policy: The student is expected to review and follow the Southwest Technical College Internet Safety Policy at: <https://stech.edu/students/policies/>

Student Code of Conduct Policy: The student is expected to review and follow the Southwest Technical College Student Code of Conduct Policy at: <https://stech.edu/students/policies/>

Accommodations: Students with medical, psychological, learning, or other disabilities desiring accommodations or services under ADA, must contact the Student Services Office. Student Services determines eligibility for and authorizes the provision of these accommodations and services. Students must voluntarily disclose that they have a disability, request an accommodation, and provide documentation of their disability. Students with disabilities may apply for accommodations, based on an eligible disability, through the Student Services office located at 757 W. 800 S., Cedar City, UT 84720, and by phone at (435) 586-2899. No diagnostic services are currently available through Southwest Technical College.

Safety and Building Maintenance: The College has developed and follows a variety of plans to ensure the safe and effective operation of its facilities and programs. The following plans are available online: 1) Facilities Operations and Maintenance Plan; 2) Technical Infrastructure Plan; and 3) Health and Safety Plan.

Withdrawals and Refunds: Please refer to the Southwest Technical College Refund Policy at: <https://stech.edu/students/policies/>

Any high school or adult student, who declares a technical training objective is eligible for admission at Southwest Technical College (Southwest Tech). Program-specific admissions requirements may exist and will be listed on the Southwest Tech website. A high school diploma or equivalent is not required for admission but is mandatory for students seeking Title IV Federal Financial Aid.

Non-Discriminatory Policy: Southwest Technical College affirms its commitment to promote the goals of fairness and equity in all aspects of the educational enterprise, and bases its policies on the idea of global human dignity.

Southwest Tech is committed to a policy of nondiscrimination. No otherwise qualified person may be excluded from participation in or be subjected to discrimination in any course, program or activity because of race, age, color, religion, sex, pregnancy, national origin or disability. Southwest Technical College does not discriminate on the basis of sex in the education programs or activities that it operates, as required by Title IX and 34 CFR part 106. The requirement not to discriminate in education programs or activities extends to admission and employment. Inquiries about Title IX and its regulations to STECH may be referred to the Title IX Coordinator, to the Department of Education, and/or to the Office for Civil rights.

If you believe you have experienced discrimination or harassment on our campus, please contact the Title IX Coordinator, Cory Estes: cestes@stech.edu, (435) 865-3938.

For special accommodations, please contact the ADA Coordinator, Cyndie Tracy: ctracy@stech.edu, (435) 865-3944.

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